

# YZV202E Optimization for Data Science Project Proposal Optimization on Electric Vehicle Charging Capacity Allocation in Istanbul

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## I. PROJECT DESCRIPTION

This project aims to optimize the electric vehicle charging infrastructure in Istanbul by improving the allocation of charging capacity across the city. The study considers both expanding existing charging stations and opening new stations at candidate locations. Since electric vehicle usage is expected to increase over time, charging demand may become unevenly distributed across different districts.

The main goal is to decide how much additional capacity should be assigned to existing stations and which candidate locations should be opened, while considering accessibility, investment cost, unmet demand, and station congestion. In the initial version, demand zones will be defined at the district level, and future demand will be estimated using population-based demand weights and a year-based electric vehicle growth ratio.

The final system will include a year slider and a demand scaling slider for scenario analysis. Optimization results will be visualized on a map, where station utilization will be shown with red circles of different sizes and newly selected stations will be marked with X symbols.

## II. PROBLEM DEFINITION

We formulate the EV charging infrastructure planning problem for Istanbul as a capacitated location assignment and capacity expansion problem. Let  $D$  denote the set of demand zones,  $E$  the set of existing charging stations, and  $C$  the set of candidate locations for new stations. We define  $S = E \cup C$  as the set of all possible charging service points.

For a selected year  $y$ , the EV growth factor is represented by a ratio  $r_y$ , which compares the EV level in year  $y$  with a selected base year.  $N_{base}$  denotes the total number of registered EVs in the base year (2022).  $r_y$  is estimated by fitting a logistic growth curve to historical TÜİK EV registration data,

and extrapolating to year  $y$  relative to the base year [5]. The estimated total EV demand for year  $y$  is defined as

$$N(y) = N_{base} r_y.$$

To account for uncertainty, a demand scaling parameter  $\theta$  is introduced through the demand range slider. Since registered EV counts and charging station capacity are not directly in the same unit, a conversion factor  $\rho$  is used to convert the estimated number of EVs into charging demand units. The charging demand of demand zone  $d$  is defined as

$$q_d(y, \theta) = \rho \theta N_{base} r_y w_d,$$

where  $w_d$  is the normalized demand weight of demand zone  $d$  and  $\sum_{d \in D} w_d = 1$ .

### Parameters

- $C^{\max} = \sum_{d \in D} q_d \cdot \max_{s \in S} c_{ds}$ : maximum possible total accessibility cost, used for normalization
- $K_i^{\max}$ ,  $\forall i \in E$ : maximum allowable capacity of existing station  $i$
- $K_j^{\max}$ ,  $\forall j \in C$ : maximum allowable capacity of candidate station  $j$
- $K_i$ ,  $\forall i \in E$ : current capacity of existing station  $i$
- $B$ : total available investment budget
- $d_{\max}$ : maximum allowable assignment distance
- $M$ : a sufficiently large constant
- $c_{ds}$ : Haversine distance between demand zone  $d$  and service point  $s$
- $g_i$ : marginal cost of adding one capacity unit to existing station  $i$  [6], [7]
- $f_j$ : fixed opening cost of candidate station  $j$  [6], [7]
- $h_j$ : marginal cost of installing one capacity unit at candidate station  $j$  [6], [7]
- $w_d$ : normalized demand weight of demand zone  $d$
- $r_y$ : year-based EV growth ratio
- $\theta$ : demand scaling factor used for scenario analysis

- $\rho$ : conversion factor from registered EV count to charging demand units

#### Decision Variables

- $x_i \geq 0, \forall i \in E$ : additional capacity assigned to existing station  $i$
- $y_j \in \{0, 1\}, \forall j \in C$ : equals 1 if candidate station  $j$  is opened, and 0 otherwise
- $z_j \geq 0, \forall j \in C$ : installed capacity at candidate station  $j$
- $a_{ds} \geq 0, \forall d \in D, s \in S$ : amount of demand from zone  $d$  assigned to service point  $s$
- $u_d \geq 0, \forall d \in D$ : unmet demand in demand zone  $d$
- $o_s \geq 0, \forall s \in S$ : overload amount at service point  $s$ , where  $o_s = o_i$  if  $s \in E$  and  $o_s = o_j$  if  $s \in C$

**Objective Function.** The objective is to minimize the weighted sum of accessibility cost, unmet demand, investment cost, and overload penalty. Overload  $o_s$  represents the amount by which demand assigned to station  $s$  exceeds its available capacity, acting as a soft constraint that allows infeasibility to be penalized rather than eliminated, improving model tractability:

$$\min \left[ \alpha \frac{\sum_{d \in D} \sum_{s \in S} c_{ds} a_{ds}}{C_{\max}} + \beta \frac{\sum_{d \in D} u_d}{\sum_d q_d} + \gamma \frac{\sum_{i \in E} g_i x_i + \sum_{j \in C} f_j y_j + \sum_{j \in C} h_j z_j}{B} + \lambda \frac{\sum_{s \in S} o_s}{\sum_d q_d} \right] \quad (1)$$

Here, Haversine distance is used as a tractable proxy for accessibility in the initial version of the project. Future work may replace this proxy with road-network travel time.

Each term is normalized by its theoretical maximum before weighting: accessibility by the maximum possible assignment cost, unmet demand and overload by total demand  $\sum_d q_d$ , and investment cost by the total budget  $B$ . This ensures the weights  $\alpha, \beta, \gamma, \lambda$  are dimensionally consistent and comparable.

**Constraints.** *Demand satisfaction:*  $\sum_{s \in S} a_{ds} + u_d = q_d(y, \theta), \forall d \in D$ . This constraint ensures that the demand of each zone is either assigned to charging service points or remains unmet.

*Capacity constraints for existing stations:*  $\sum_{d \in D} a_{di} \leq K_i + x_i + o_i, \forall i \in E$ .

*Expansion limit for existing stations:*  $x_i \leq K_i^{\max} - K_i, \forall i \in E$ . Expansion of existing stations is bounded by a realistic maximum capacity  $K_i^{\max}$ , estimated from physical site constraints.

*Capacity constraints for candidate stations:*  $\sum_{d \in D} a_{dj} \leq z_j + o_j, \forall j \in C$ .

*Opening and capacity limit constraints for candidate stations:*  $z_j \leq K_j^{\max} y_j, \forall j \in C$  and  $\sum_{d \in D} a_{dj} \leq M y_j, \forall j \in C$ . These ensure that a candidate station can receive installed capacity and serve demand only if it is opened, while also bounding its installed capacity by a realistic maximum capacity  $K_j^{\max}$ .

*Budget constraint:*  $\sum_{i \in E} g_i x_i + \sum_{j \in C} f_j y_j + \sum_{j \in C} h_j z_j \leq B$ . This constraint ensures that the total investment cost does not exceed the available budget.

*Maximum distance constraint:*  $a_{ds} = 0$  if  $c_{ds} > d_{\max}, \forall d \in D, s \in S$ . This constraint prevents demand from being assigned to service points that are farther than the maximum allowable Haversine distance.

*Non-negativity constraints:*  $x_i, z_j, a_{ds}, u_d, o_s \geq 0$ .

### III. DATASET

The project will use open datasets related to Istanbul's EV charging infrastructure, parking locations, and demographic distribution. In the initial version, demand zones will be defined at the district level. Each district will be represented by its geographic centroid, and charging demand will be assigned from these centroids to existing or candidate charging service points.

The existing EV charging station dataset [1] will be used to construct the set of existing stations  $E$  and to obtain station coordinates. The charging socket dataset [2] will be used to estimate the current capacity parameter  $K_i$  for each existing station. ISPARK parking lot locations [3] will be used as candidate locations  $C$  for possible new charging stations. District-level demographic data [4] will be used to construct demand weights  $w_d$  by normalizing the population of each district.

The data engineering process will consist of the following steps. First, all spatial datasets will be cleaned and converted into a common coordinate format. Second, existing charging stations and socket records will be matched to estimate current station capacity. Third, district-level population values will be normalized to obtain demand weights. Fourth, EV growth ratios  $r_y$  will be computed from official vehicle statistics [5] and used with the conversion factor  $\rho$  to scale the base demand level into charging demand units. Finally, a distance matrix  $c_{ds}$  will be computed using Haversine distance between district centroids and charging service point coordinates.

The resulting model inputs will be the demand zones  $D$ , existing stations  $E$ , candidate locations  $C$ , current capacities  $K_i$ , candidate capacity limits  $K_j^{\max}$ , demand weights  $w_d$ , year-based growth ratios  $r_y$ , the conversion factor  $\rho$ , and the accessibility cost matrix  $c_{ds}$ .

### IV. METHODOLOGY

The proposed problem will be modeled as a mixed-integer linear programming problem. The binary decision variables  $y_j$  determine whether candidate stations are opened, while the continuous decision variables determine capacity expansion, installed capacity, demand assignment, unmet demand, and overload levels.

The methodology consists of five main stages. First, the raw datasets will be cleaned and transformed into model inputs. Existing charging station and socket data will be used to construct the current infrastructure and estimate existing capacities. Parking lot data will be used to define candidate

locations, and district-level demographic data will be used to define demand zones and demand weights.

Second, year-based EV demand will be estimated using the growth ratio  $r_y$  and converted into charging demand units using  $\rho$ . Instead of directly relying on absolute EV numbers, the model will use the ratio between the selected year and a base year. Specifically,  $r_y$  is computed from TÜİK annual EV registration records by fitting a logistic growth curve to historical data. This allows the model to extrapolate demand for future years in a methodologically consistent manner, while remaining suitable for scenario analysis.

Third, the accessibility cost matrix will be constructed by calculating Haversine distances between demand zone centroids and charging service points. Haversine distance is used as a tractable approximation of accessibility in the initial version. In future work, this can be replaced with road-network travel time.

Fourth, the MILP model and a greedy heuristic are used to solve the problem. The greedy heuristic iteratively opens candidate stations in order of cost-effectiveness (demand covered per unit investment cost) until the budget is exhausted, providing a fast approximate solution for comparison with the MILP optimum.

Finally, the results will be visualized on an interactive map. Station utilization levels will be shown with red circles sized proportionally to utilization, and newly opened candidate stations will be marked with X symbols. Interactive sliders allow users to adjust the year, demand scaling parameter  $\theta$ , and normalized weights  $\alpha, \beta, \gamma, \lambda$  for scenario analysis. Sensitivity analysis on  $\lambda$  (overload penalty) is conducted to evaluate its impact on solution structure.

## REFERENCES

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## V. GITHUB REPOSITORY

Project Repository (Click)